Linear Models Lecture 16: GMM Inference and Going Beyond ATE

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Recap: The GMM Framework

From Lecture 15:

The GMM estimator minimizes the weighted quadratic form:

$$\hat{\beta}_{\text{gmm}} = \arg\min_{\beta} \; n\,\bar{g}_n(\beta)' W \,\bar{g}_n(\beta)$$

Key results:

- $W = (Z'Z)^{-1}$ gives 2SLS (Thm 13.2)
- Sandwich variance: $V_\beta = (Q'WQ)^{-1}(Q'W\Omega WQ)(Q'WQ)^{-1}$ (Thm 13.3)
- Efficient GMM: $W = \Omega^{-1}$, $V_\beta = (Q'\Omega^{-1}Q)^{-1}$ (Thm 13.4)
- 2SLS efficient only under homoskedasticity (Thm 13.6)
- Two-step and iterated GMM are asymptotically efficient (Thm 13.7)

Today: What can we *test* with GMM? And how does GMM let us go *beyond* average treatment effects?

The Overidentification Test: Intuition

Goal: a specification test of our identifying assumptions. We assumed $\mathbb{E}[Z_i e_i] = 0$. If *any* of those moments is wrong (instrument correlates with e, wrong functional form, bad exclusion), GMM is inconsistent — we want a check.

Why overidentification gives leverage.

- $\ell = k$ (just identified): FOC sets $\bar{g}_n(\hat{\beta}) = 0$ exactly — nothing to test.
- $\ell > k$: only k linear combinations of $\bar{g}_n$ are pinned to zero by the FOC. The remaining $\ell - k$ are *free* to disagree with zero — that is what we test.

Test logic:

- **Small** $J(\hat{\beta})$: moments jointly compatible $\Rightarrow$ no evidence against the model
- **Large** $J(\hat{\beta})$: no β satisfies all moments at once $\Rightarrow$ misspecification

$$H_0 : \mathbb{E}[Ze] = 0 \qquad \text{vs.} \qquad H_1 : \mathbb{E}[Ze] \neq 0$$

Hansen's J-Test (Hansen Thm. 13.14)

Theorem (13.14: Overidentification Test)

Under $H_0 : \mathbb{E}[Ze] = 0$ *and using an efficient weight matrix estimator,*

$$J = J(\hat{\beta}_{gmm}) = n\,\bar{g}_n(\hat{\beta})'\hat{\Omega}^{-1}\bar{g}_n(\hat{\beta}) \xrightarrow{d} \chi^2_{\ell-k}$$

Reject H_0 *if* $J > \chi^2_{\ell-k,1-\alpha}$.

Notes:

- Degrees of freedom $= \ell - k =$ number of overidentifying restrictions
- Generalizes the Sargan test (which assumed homoskedasticity)
- When $\ell = k$ (just identified): $J = 0$ always — no test possible

Why the Efficient Weight $\hat{\Omega}^{-1}$?

The choice of weight matrix is what makes J a clean χ^2.

Under H_0, the sample moments are asymptotically normal:

$$\sqrt{n}\,\bar{g}_n(\hat{\beta}) \xrightarrow{d} \mathcal{N}(0,\ V_g), \qquad V_g \propto \Omega.$$

Standardizing. Recall: if $X \sim \mathcal{N}(0, \Sigma)$, then $X'\Sigma^{-1}X \sim \chi^2$. The inverse-variance weight is what converts a Gaussian vector into a χ^2 scalar.

In GMM. Weighting $\bar{g}_n$ by $\hat{\Omega}^{-1}$ does the same job — it standardizes the moments by their (inverse) variance. With any other $\hat{W}$,

$$J = n\,\bar{g}_n'\hat{W}\bar{g}_n \xrightarrow{d} \sum_{j=1}^{\ell-k} \lambda_j \chi^2_{1,j},$$

a *weighted sum* of χ^2's with unknown coefficients λ_j. No clean reference distribution.

J-Test: Strengths and Limitations

Strengths:

- Automatic byproduct of efficient GMM estimation
- General: works under heteroskedasticity, no distributional assumptions
- Natural diagnostic: "always report the J statistic" (Hansen, Ch. 13)

Limitations:

- No power if all instruments are invalid in the same direction — moments are all "wrong" together, J-test cannot detect this
- Requires $\ell > k$ (overidentification)
- Rejection tells you *something* is wrong, but not *what*

Good practice: Use subset overidentification tests (Thm 13.15) to investigate *which* instruments may be invalid.

J-Test for Missing Data

From the Abrevaya & Donald application (Lecture 15):

```
# Run the specification test
specTest(gmm_men)

# Hansen's J-test
# Test E(g) = 0:
# Statistics   df   p-value
# J-test        ?    2     ?
```

Interpretation ($\ell - k = 7 - 5 = 2$ d.f.):

- Tests whether the linear projection $x = z'\gamma + \xi$ holds for both observed and missing cases
- **Fail to reject:** the projection is stable across observed/missing
- **Reject:** missingness depends on unobservables, violating Assumption 1

From Specification to Parameter Restrictions

The J-test asked: *is the model correctly specified?* ($H_0 : \mathbb{E}[g_i(\beta)] = 0$.)

We now ask a different question: *given the model, does a particular parameter restriction hold?*

$$H_0 : r(\beta) = \theta_0, \qquad r : \mathbb{R}^k \to \mathbb{R}^q$$

e.g. $\beta_2 = 0$, $\beta_2 = \beta_3$, $\beta_2/\beta_3 = 1$, or a nonlinear policy quantity.

GMM gives two natural statistics for this question (a third, the score / LM test, we won't cover):

- **Wald** (Thm 13.8) — uses the unrestricted estimator only; familiar from t/F, now nonlinear-capable
- **Distance** (Thm 13.12) — compares restricted vs. unrestricted criterion values; the GMM analogue of the LR test, invariant to reparameterization

The Wald Test (Hansen Thm. 13.8)

We have used t/F-statistics since Lecture 1. The Wald test is the GMM generalization — now r can be *nonlinear*, and $\hat{V}_\beta$ is the sandwich (no homoskedasticity assumption).

Test $H_0 : \theta = \theta_0$ where $\theta = r(\beta)$.

Theorem (13.8: Wald Test)

Under H_0, as $n \to \infty$:

$$W = n(\hat{\theta} - \theta_0)' \hat{V}_\theta^{-1} (\hat{\theta} - \theta_0) \xrightarrow{d} \chi^2_q$$

where $\hat{V}_\theta = \hat{R}' \hat{V}_\beta \hat{R}$ and $\hat{R} = \frac{\partial}{\partial \beta} r(\hat{\beta}_{gmm})'$.

Familiar special case: Linear r with $H_0 : \beta_j = 0$ gives $W = \hat{\beta}_j^2 / \widehat{\text{Var}}(\hat{\beta}_j) = t_j^2 \xrightarrow{d} \chi^2_1$ — the squared t-statistic.

The Distance Test (Hansen Thm. 13.12)

An alternative to the Wald test, based on comparing criterion functions.

Idea: Estimate unrestricted ($\hat{J}$) and restricted ($\tilde{J}$, subject to $r(\beta) = \theta_0$) models by efficient GMM.

Theorem (13.12: Distance Test)

Under H_0 and using efficient GMM, as $n \to \infty$:

$$D = \tilde{J} - \hat{J} \xrightarrow{d} \chi^2_q$$

Key advantages over Wald:

- Invariant to reparameterization—the Wald statistic is not
- Analogous to the likelihood ratio test (criterion-based)

When Does the Distance Test Agree with Wald? (Thm 13.13)

When the same weight matrix $\hat{\Omega}^{-1}$ is used for both restricted and unrestricted fits:

- $D \geq 0$ always — the restricted minimum cannot fall below the unrestricted one (it is the same objective, optimized over a smaller set).
- For *linear* restrictions $r(\beta) = R\beta - \theta_0$, $D = W$ exactly: the two tests deliver the same number.

Consequence. Wald and Distance can disagree only when r is nonlinear — and there, the Distance test is preferred (invariant).

Why $D = W$ in the linear case? We work through it over the next four slides: efficient minimum distance (Hansen §8.25), the Lagrange solution, the specialization to CGMM (Thm 13.19), and the proof.

Efficient Restricted Estimation (Hansen §8.25)

Problem. We have an unrestricted estimator $\hat{\beta}$ with asymptotic variance V. We want the most efficient estimator that imposes a linear constraint $R'\beta = c$.

Hansen's answer. Take the point closest to $\hat{\beta}$ in the metric defined by V^{-1}, subject to the constraint:

$$\tilde{\beta} = \arg \min_{\beta \,:\, R'\beta = c} (\hat{\beta} - \beta)' V^{-1} (\hat{\beta} - \beta).$$

Why this metric? V^{-1} standardizes the deviations of $\hat{\beta}$ component by component. Directions in which $\hat{\beta}$ is imprecise get downweighted; directions in which it is precise get heavy weight. The Mahalanobis distance is the GLS analog of squared distance, and it is optimal precisely because it accounts for the precision of $\hat{\beta}$.

Geometric reading. $\tilde{\beta}$ is the orthogonal projection of $\hat{\beta}$ onto the constraint set $\{\beta : R'\beta = c\}$, with orthogonality measured in the V^{-1}-inner-product.

Solving the Minimum Distance Problem

Lagrangian.

$$L \;=\; (\hat{\beta}-\beta)'V^{-1}(\hat{\beta}-\beta) + 2\lambda'(R'\beta - c).$$

First-order conditions.

- In β: $-2V^{-1}(\hat{\beta}-\beta) + 2R\lambda = 0$, so $\hat{\beta}-\beta = VR\lambda$.
- In λ: $R'\beta = c$.

Solve for λ. Premultiply the first FOC by R' and apply the constraint:

$$R'\hat{\beta} - c \;=\; R'VR\,\lambda \quad\Longrightarrow\quad \lambda \;=\; (R'VR)^{-1}(R'\hat{\beta}-c).$$

Closed form. Substituting λ back,

$$\boxed{\tilde{\beta} \;=\; \hat{\beta} \;-\; VR\,(R'VR)^{-1}(R'\hat{\beta}-c).}$$

The correction is a V-weighted projection of the constraint violation $R'\hat{\beta}-c$ back onto the parameter space.

From Minimum Distance to CGMM (Theorem 13.19)

The Chapter 8 formula gives the efficient restricted estimator for *any* $\hat{\beta}$ with variance V. To get the constrained GMM estimator, plug in the efficient GMM variance.

Specialization. Take

$$V = \hat{V}_\beta = (\hat{Q}'\hat{\Omega}^{-1}\hat{Q})^{-1} \quad \text{(Theorem 13.11).}$$

Out comes Hansen 13.19.

$$\boxed{\hat{\beta}_{\text{cgmm}} = \hat{\beta}_{\text{gmm}} - \hat{V}_\beta R\,(R'\hat{V}_\beta R)^{-1}(R'\hat{\beta}_{\text{gmm}} - c).}$$

Why this is the right plug-in. Efficient GMM is asymptotically a GLS estimator: it minimizes a quadratic form in the moments with weight $\hat{\Omega}^{-1}$, equivalent to minimizing a quadratic form in $\beta - \beta_0$ with weight $\hat{V}_\beta^{-1}$. Imposing $R'\beta = c$ efficiently is just the constrained version of the same quadratic problem. The minimum-distance step and the GMM step share the metric $\hat{V}_\beta^{-1}$, which is why the two formulas coincide.

Proof of $D = W$ for Linear r

Two ingredients.

- Variance: $\hat{V}_\beta = (\hat{Q}'\hat{\Omega}^{-1}\hat{Q})^{-1}$ (Thm 13.11).
- Constrained shift: $\hat{\beta}_{\text{cgmm}} - \hat{\beta}_{\text{gmm}} = -\hat{V}_\beta R\,(R'\hat{V}_\beta R)^{-1}u$, where $u := R'\hat{\beta}_{\text{gmm}} - c$.

Step 1: expand J around $\hat{\beta}_{\text{gmm}}$. The unrestricted FOC is $\partial J/\partial\beta\big|_{\hat{\beta}_{\text{gmm}}} = 0$, so the linear term in the Taylor expansion drops. For linear moments the quadratic expansion is exact:

$$D \;=\; J(\hat{\beta}_{\text{cgmm}}) - J(\hat{\beta}_{\text{gmm}}) \;=\; n\,(\hat{\beta}_{\text{cgmm}} - \hat{\beta}_{\text{gmm}})'\,\hat{V}_\beta^{-1}(\hat{\beta}_{\text{cgmm}} - \hat{\beta}_{\text{gmm}}).$$

Step 2: substitute the constrained shift. The middle factor $\hat{V}_\beta\,\hat{V}_\beta^{-1}\,\hat{V}_\beta$ collapses to $\hat{V}_\beta$:

$$D \;=\; n\,u'\,(R'\hat{V}_\beta R)^{-1}(R'\hat{V}_\beta R)(R'\hat{V}_\beta R)^{-1}u \;=\; n\,u'\,(R'\hat{V}_\beta R)^{-1}u.$$

Step 3: match the Wald statistic. The Wald test uses $\hat{V}_\theta = R'\hat{V}_\beta R$, so

$$D \;=\; n\,u'\,\hat{V}_\theta^{-1}u \;=\; W. \qquad \blacksquare$$

Three Tests Compared

	Wald Test	Distance Test	J-Test
Tests	Parameter restrictions $r(\beta) = \theta_0$	Parameter restrictions $r(\beta) = \theta_0$	Model specification $\mathbb{E}[g_i(\beta)] = 0$
Requires	Unrestricted estimate only	Both restricted and unrestricted	Efficient weight matrix
Distribution	χ^2_q	χ^2_q	$\chi^2_{\ell-k}$
Invariant?	No	Yes	N/A
Analogy	t-test / F-test	Likelihood ratio	Sargan test

Recommendation: Distance test for nonlinear hypotheses (invariant). Wald test for linear hypotheses (simpler, equivalent). Always report the J-test.

Subset Overidentification Tests (Hansen Thm. 13.15)

Question: Are *specific* instruments valid?

Partition $Z = (Z_a, Z_b)$:

- Z_a (ℓ_a instruments): believed valid, $\mathbb{E}[Z_a e] = 0$
- Z_b (ℓ_b instruments): questionable, $\mathbb{E}[Z_b e] \stackrel{?}{=} 0$

Require $\ell_a > k$ so Z_a alone identifies the model.

Theorem (13.15: Subset Overidentification Test)

Let $\tilde{J}$ use only Z_a and $\hat{J}$ use all $Z = (Z_a, Z_b)$. Under $H_0 : \mathbb{E}[Z_b e] = 0$,

$$C = \hat{J} - \tilde{J} \xrightarrow{d} \chi^2_{\ell_b}.$$

Special case: If Z_a is just-identified ($\ell_a = k$), then $\tilde{J} = 0$ and $C = \hat{J}$ is the standard J-test.

Endogeneity Test via GMM (Hansen Thm. 13.16)

Question: Is Y_2 endogenous, i.e., is $\mathbb{E}[Y_2 e] \neq 0$?

Model: $Y = Z_1'\beta_1 + Y_2'\beta_2 + e$, instruments (Z_1, Z_2) with $\mathbb{E}[Z_1 e] = \mathbb{E}[Z_2 e] = 0$.

Key insight: If Y_2 is exogenous (H_0), then Y_2 is a valid instrument for itself. So we can expand the instrument set from (Z_1, Z_2) to (Z_1, Z_2, Y_2).

Theorem (13.16: Endogeneity Test)

1. *Estimate efficient GMM with instruments (Z_1, Z_2). Let $\bar{J}$ be the criterion.*
2. *Estimate efficient GMM with instruments (Z_1, Z_2, Y_2). Let $\hat{J}$ be the criterion.*

Under $H_0 : \mathbb{E}[Y_2 e] = 0, \quad C = \hat{J} - \bar{J} \xrightarrow{d} \chi^2_{k_2}.$

This is a **subset overidentification test** where the "questionable instruments" are Y_2 itself. It **generalizes** Durbin-Wu-Hausman: DWH requires homoskedasticity; this GMM version does not.

GMM Subsumes All Prior Tests

Earlier Course Test	**GMM Version**	**Theorem**
t-test for $\beta_j = 0$	Wald test ($q = 1$)	13.8
F-test for $R\beta = c$	Wald test ($q > 1$)	13.8
Hausman (OLS vs IV)	Endogeneity test	13.16
Durbin-Wu-Hausman	Endogeneity test	13.16
Sargan overid test	J-test	13.14
White's heteroskedasticity test	Special case of J-test	—
Likelihood ratio test	Distance test	13.12

Meta-lesson #3 (General): GMM provides a *unified* framework for both estimation and inference. Every test we have seen is a special case of a GMM test—often under weaker assumptions.

When ATE Is Not Enough

Recall from Lecture 14:

- **LATE** = effect for *compliers* only — those induced to change treatment by the instrument
- Different instruments ⇒ different complier groups ⇒ different LATEs

Key questions that LATE cannot answer:

1. What is the effect for *always-takers*? For *never-takers*?
2. Does the treatment effect *change sign* across subpopulations?
3. What would happen under a *different policy* (different compliance margin)?

> The **Marginal Treatment Effect** $\Delta^{MTE}(u_D)$ traces out treatment effects across the *entire* resistance-to-treatment distribution. Estimating MTE requires GMM.

West German TV in East Germany

Kern & Hainmueller (2009): Effect of Western media on support for the East German regime.

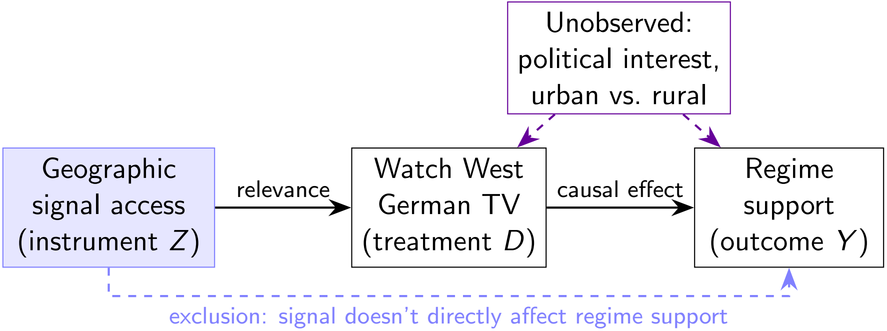
Setting:

- Cold War East Germany (GDR), pre-1989
- West German TV broadcast entertainment, news, and consumer culture
- The regime feared Western media as subversive
- Most East Germans could receive West German TV signals

Research question: Did exposure to West German TV *undermine* support for the Communist regime?

Outcome: Support for the regime (survey data)
Treatment: Regular viewing of West German TV
Challenge: Viewers self-select into watching — endogenous

The Instrument: Geographic Signal Access



Key: Due to topography and transmitter locations, some areas of East Germany could not receive West German TV — notably the **Dresden** region ("Valley of the Clueless").

Geographic signal access is plausibly exogenous: determined by physics, not politics.

Conventional IV Results

Standard IV/2SLS estimate:

$$\widehat{LATE} \approx -0.12$$

Interpretation: Among *compliers* (those induced to watch by having signal access), watching West German TV reduces regime support by 0.12 standard deviations.

But who are the compliers?

- People who watch West German TV *only when they have signal access*
- Not the most politically interested (they would find other sources — always-takers)
- Not the most regime-loyal (they would not watch even with access — never-takers)
- Compliers are a potentially narrow and unrepresentative group

Question: What about the effects for always-takers and never-takers?

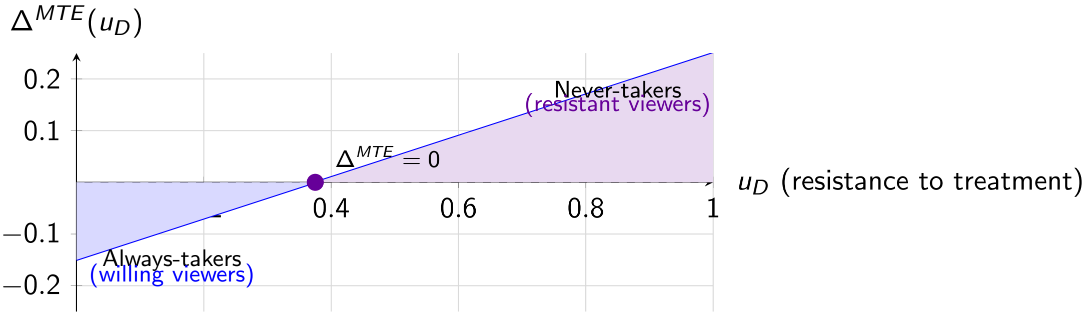
MTE Results: Treatment Effect Heterogeneity

Using the MTE framework (extending Kern & Hainmueller with MTE methods):

Group	Treatment Effect	Interpretation
Always-takers	-0.104	Western TV *reduces* support (politically interested viewers)
Compliers (LATE)	≈ -0.12	Moderate effect
Never-takers	$+0.189$	Western TV *increases* support! (contrast effect / reactance)

Sign reversal: The treatment effect is *negative* for willing viewers but *positive* for resistant viewers! Western consumer culture made regime-loyal individuals *more* supportive (reactance), while curious viewers became *less* supportive (information effect).

Visualizing the MTE Curve



Low u_D: Always-takers (willing viewers) — effect is *negative*. **High u_D:** Never-takers (resistant viewers) — effect is *positive*. The **LATE** is a weighted average over compliers, masking the heterogeneity.

Why MTE Requires GMM

MTE estimation involves:

1. **Nonlinear moment conditions:** $E[Y|X,Z] = X'\beta + \int_0^{P(Z)} \Delta^{MTE}(u)\, du$. Parameters enter through the integral — nonlinear in β.
2. **Overidentification:** Multiple instruments provide more moment conditions than parameters $\Rightarrow$ testable.
3. **Efficient weighting:** Different propensity score regions are estimated with different precision $\Rightarrow$ optimal weighting matters.
4. **J-test:** Tests whether the MTE specification (e.g., polynomial degree) is adequate.

Without GMM, we could not estimate or test the MTE.

Policy Implications

The **Policy-Relevant Treatment Effect** (PRTE) depends on which part of the MTE curve the policy targets:

$$PRTE = \int_0^1 \Delta^{MTE}(u_D) \cdot \omega^{PRTE}(u_D)\, du_D$$

where ω^{PRTE} depends on how the policy shifts the propensity score.

For the propaganda example:

- Forcing everyone to watch (shifting never-takers) could backfire: MTE is positive for high u_D
- Facilitating access for willing viewers (low u_D) would reduce regime support
- The PRTE sign/magnitude depends on the policy's compliance margin

Lesson: Treatment effect heterogeneity means policy evaluation requires more than a single number. The MTE curve—estimated via GMM—provides the necessary detail.

Connection to ivmte

Recall from Lecture 14: the `ivmte` package (Shea & Torgovitsky) estimates MTE:

```
library(ivmte)
result <- ivmte(
  data = df,
  outcome = "regime_support",
  treatment = "watch_western_tv",
  instrument = "signal_access",
  target = "ate",        # or "att", "prte"
  m0 = ~ u + I(u^2),    # MTE polynomial for control
  m1 = ~ u + I(u^2),    # MTE polynomial for treated
  propensity = D ~ signal_access
)
```

Under the hood, `ivmte` **performs:** nonlinear GMM estimation of MTE parameters, efficient weighting across moment conditions, and overidentification testing (J-test).

Restricted GMM (Hansen §13.15–13.16)

Linear constraints: $R'\beta = c$

$\hat{\beta}_{\text{cgmm}} = \hat{\beta}_{\text{gmm}} - (X'ZWZ'X)^{-1}R\left(R'(X'ZWZ'X)^{-1}R\right)^{-1}(R'\hat{\beta}_{\text{gmm}} - c)$

This is the GMM analog of restricted OLS / minimum distance.

Nonlinear constraints: $r(\beta) = 0$ — minimize $J(\beta)$ subject to constraint (numerical optimization).

Theorem (13.10–13.11)

The constrained efficient GMM estimator has asymptotic variance $V_{cgmm} = V_\beta - V_\beta R(R'V_\beta R)^{-1}R'V_\beta$. Constrained estimation is (weakly) more efficient than unconstrained — if the constraints are true.

The Bootstrap: The Idea (Efron, 1979)

The sampling distribution of $\hat{\theta}$ depends on the unknown population CDF F. With F in hand, exact p-values and CIs are immediate. We do not have F.

Efron's substitution. Replace F with the empirical CDF

$$\hat{F}_n(t) = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}\{X_i \leq t\},$$

which places mass $1/n$ on each observed datum.

Plug-in / bootstrap principle. For any functional $T(\cdot)$, the bootstrap analog is $T(\hat{F}_n)$. For sampling distributions,

$$\mathcal{L}\left(\hat{\theta} - \theta_0 \mid F\right) \approx \mathcal{L}\left(\hat{\theta}^* - \hat{\theta} \mid \hat{F}_n\right).$$

By Glivenko–Cantelli, $\hat{F}_n \to F$ uniformly. For sufficiently smooth functionals the approximation is therefore consistent: real-world inference reduces to Monte Carlo inside the sample.

The Bootstrap: Algorithm

Nonparametric (pairs) bootstrap.

1. Compute $\hat{\theta}$ on the original sample $\{(Y_i, X_i, Z_i)\}_{i=1}^{n}$.
2. For $b = 1, \ldots, B$:
   - Draw $\{(Y_i^*, X_i^*, Z_i^*)\}_{i=1}^{n}$ with replacement from the original sample.
   - Recompute $\hat{\theta}_b^*$ on the resample.
3. Use the empirical distribution of $\hat{\theta}_b^* - \hat{\theta}$ as a proxy for the distribution of $\hat{\theta} - \theta_0$.

What you can read off the resamples:

- Bootstrap SE: $\widehat{\mathsf{SE}}_B(\hat{\theta}) = \mathrm{sd}(\hat{\theta}_b^*)$.
- Percentile CI: empirical quantiles of $\{\hat{\theta}_b^*\}$.
- Percentile-t CI: empirical quantiles of $t_b^* = (\hat{\theta}_b^* - \hat{\theta})/\widehat{\mathsf{SE}}(\hat{\theta}_b^*)$.

How big should B be? For SEs, $B \approx 200$ is fine; for two-sided CIs, $B \geq 999$; for tail p-values, $B \geq 4{,}999$.

Why Bootstrap? Asymptotic Refinement

If V_β and the standard normal already give us a CI, why bother simulating?

Edgeworth expansion. For a smooth pivotal statistic $t = (\hat{\theta} - \theta_0)/\widehat{\mathrm{SE}}$,

$$\Pr(t \leq x) = \Phi(x) + \frac{p_1(x)\phi(x)}{\sqrt{n}} + \frac{p_2(x)\phi(x)}{n} + O(n^{-3/2}).$$

The normal approximation drops every correction term: error is $O(n^{-1/2})$.

What the bootstrap captures:

- Bootstrapping $\hat{\theta}^*$ (percentile method) matches only $\Phi(x)$: error remains $O(n^{-1/2})$.
- Bootstrapping the pivot t_b^* matches the $1/\sqrt{n}$ term as well: error drops to $O(n^{-1})$.

Takeaway from the expansion. Refinement requires bootstrapping a *pivot*, i.e. a statistic whose limiting distribution is free of nuisance parameters. Studentize first, then bootstrap.

Bootstrap Variants: Match the DGP

- **Nonparametric / pairs.** Resample rows of the data. Robust to misspecified conditional moments. Default for i.i.d. cross-sections.
- **Parametric.** Simulate Y_i^* from $F_{\hat{\theta}}$, e.g. $\mathcal{N}(X_i'\hat{\beta}, \hat{\sigma}^2)$. Tighter when the parametric model is correct, biased when it is not.
- **Residual bootstrap.** Resample residuals $\hat{e}_i$ and set $Y_i^* = X_i'\hat{\beta} + \hat{e}_i^*$. Imposes conditional homoskedasticity, so do not use under heteroskedasticity.
- **Wild bootstrap.** $Y_i^* = X_i'\hat{\beta} + v_i\hat{e}_i$ with v_i mean zero, unit variance (e.g. Rademacher ± 1). Holds X_i fixed, accommodates heteroskedasticity, works well with few clusters.
- **Cluster bootstrap.** Resample clusters and keep within-cluster observations together. Required under clustered dependence.
- **Block bootstrap.** Resample contiguous blocks. Required for time-series dependence.

Bootstrap for GMM (Hansen §13.26)

Standard bootstrap for GMM: resample (Y_i^*, X_i^*, Z_i^*) and re-estimate.

Problem: when overidentified, $\bar{g}_n = \frac{1}{n}\sum_i Z_i\hat{e}_i \neq 0$ in the original sample, so the bootstrap "population" violates $\mathbb{E}[g_i(\beta)] = 0$ at $\hat{\beta}$. The bootstrap estimator does not satisfy the orthogonality condition, and the asymptotic refinement is lost.

Solution: Recentered bootstrap (Hall & Horowitz, 1996)

$$\hat{\beta}^{**}_{\text{gmm}} = (X^{*\prime} Z^* W^* Z^{*\prime} X^*)^{-1}(X^{*\prime} Z^* W^*(Z^{*\prime} Y^* - Z'\hat{e}))$$

where $\hat{e} = Y - X\hat{\beta}_{\text{gmm}}$ comes from the *original* sample. Subtracting $Z'\hat{e}$ shifts the bootstrap moments back to mean zero at $\hat{\beta}$, restoring the refinement.

Recentered Bootstrap in R

Overidentified 2SLS, three instruments. Percentile-t CI for $\hat{\beta}_x$. To implement Hall–Horowitz, subtract an offset from y on each resample, chosen so $Z_b'\,\text{off} = \sum_i Z_i \hat{e}_i$.

```
library(AER); library(sandwich); set.seed(1)
fit   <- ivreg(y ~ x | z1 + z2 + z3, data = df)
b.h   <- coef(fit)["x"];  V.h <- vcovHC(fit, "HC1")["x","x"]
Z     <- model.matrix(~ z1 + z2 + z3, df)
shift <- crossprod(Z, resid(fit))          # sum_i Z_i e.hat_i (nonzero under overid)
n <- nrow(df);  B <- 999
boot <- function() {
  i   <- sample.int(n, n, replace = TRUE);  Zb <- Z[i, ]
  off <- as.vector(Zb %*% solve(crossprod(Zb), shift))
  db  <- df[i, ];  db$y <- db$y - off
  fb  <- ivreg(y ~ x | z1 + z2 + z3, data = db)
  (coef(fb)["x"] - b.h) / sqrt(vcovHC(fb, "HC1")["x","x"])
}
t.star <- replicate(B, boot())
ci <- b.h + c(-1, 1) * quantile(abs(t.star), 0.95) * sqrt(V.h)  # percentile-t
```

Recentered Bootstrap: Practical Advice

- **Use enough replications.** $B \geq 999$ for two-sided 95% CIs; $B \geq 4{,}999$ for tail p-values; for SEs alone, $B \approx 200$ is fine.
- **Bootstrap the studentized statistic, not the SE.** Percentile-on-$\hat{\beta}$ alone forfeits the refinement of the previous slide.
- **Always recenter when overidentified.** Dropping the `shift` term leaves the bootstrap DGP violating $\mathbb{E}[g_i(\beta)] = 0$ at $\hat{\beta}$, and the refinement is lost.
- **Refresh the weight matrix.** For two-step GMM, re-estimate $\hat{W}^*$ on each resample. Reusing $\hat{W}$ from the original sample biases coverage in small samples.
- **Match resampling to dependence.** Cluster-resample if data are clustered; block-resample for time series; wild bootstrap when clusters are few.
- **Report what you ran.** B, the resampling unit, the statistic bootstrapped, and the CI rule. Coverage claims depend on each.

The Full GMM Testing Toolkit

Test	Distribution	Purpose
J-test (Thm 13.14)	$\chi^2_{\ell-k}$	Model specification
Wald (Thm 13.8)	χ^2_q	Parameter restrictions
Distance (Thm 13.12)	χ^2_q	Parameter restrictions (invariant)
Subset overid (Thm 13.15)	$\chi^2_{\ell_b}$	Validity of specific instruments
Endogeneity (Thm 13.16)	$\chi^2_{k_2}$	Exogeneity of regressors
Constrained (Thm 13.10)	—	Efficient estimation under H_0
Bootstrap	—	Finite-sample inference

All tests: valid under heteroskedasticity, derived from the GMM criterion, applicable to linear and nonlinear models.

Semiparametric Efficiency Bound

Chamberlain (1987): If all that is known is $\mathbb{E}[g_i(\beta)] = 0$, this is a **semiparametric** problem (the distribution of the data is unknown).

Chamberlain showed that no semiparametric estimator can have asymptotic variance smaller than $(G'\Omega^{-1}G)^{-1}$ where $G = \mathbb{E}\left[\frac{\partial}{\partial\beta'} g_i(\beta)\right]$.

Efficient GMM achieves this bound: $V_\beta = (Q'\Omega^{-1}Q)^{-1} = (G'\Omega^{-1}G)^{-1}$

This means:

- No other estimator using only these moment conditions can do better
- MLE *can* do better — but requires specifying the full distribution
- GMM is the best you can do without distributional assumptions

When to Use GMM vs. Other Methods

	When to use	Advantages	Cost
OLS/GLS	$\mathbb{E}[Xe] = 0$ (no endogeneity)	Simple, efficient under assumptions	Biased if endogenous
2SLS	Endogeneity, homoskedastic	Simple, familiar	Inefficient under heterosked.
GMM	Endogeneity, heteroskedastic	Efficient, flexible, testable	More complex
MLE	Full distribution known	Most efficient (Cramér–Rao)	Misspecification bias

Decision rule: OLS/GLS if no endogeneity; 2SLS if endogenous + homoskedastic; **GMM** if endogenous + heteroskedastic or many instruments; MLE if the full distribution is known.

The Three Meta-Lessons Revisited

1. **Semiparametric**
   - Requires only moment conditions — achieves Chamberlain bound
   - Missing data: no distributional assumptions on missingness
   - MTE: no parametric model for selection — only moments from the propensity score
2. **Efficient**
   - Optimal weighting $W = \Omega^{-1}$ minimizes variance
   - Missing data: GMM has lowest MSE across methods
   - Propaganda: efficient weighting across propensity score values
3. **General**
   - Unified estimation: OLS, GLS, IV, 2SLS all as special cases
   - Unified testing: t, F, Hausman, Sargan all as special cases
   - Extends to nonlinear models, treatment effect heterogeneity, policy evaluation

The Course in One Slide

$$\begin{tikzcd} \begin{array}{c}\text{OLS}\\ E[Xe]=0\end{array} \arrow[r, "\text{heterosked.}"] & \begin{array}{c}\text{GLS}\\ E[\bar{X}e]=0\end{array} \arrow[r, "\text{endogeneity}"] & \begin{array}{c}\text{IV}\\ E[Ze]=0\\ \ell=k\end{array} \arrow[r, "\text{overid.}"] & \begin{array}{c}\text{2SLS}\\ E[Ze]=0\\ \ell>k\end{array} \arrow[r, "\text{efficiency}"] & \begin{array}{c}\textbf{GMM}\\ E[g_i(\beta)]=0\\ \textbf{optimal } W\end{array} \end{tikzcd}$$

All are moment condition estimators: $\mathbb{E}[g_i(\beta)] = 0$

The progression: Each step addresses a *new problem* (heteroskedasticity, endogeneity, overidentification, efficiency). Each estimator is a *special case* of the next. GMM is the *most general*: it nests everything and achieves the semiparametric bound.

Looking Ahead: Panel Data

Next week: Panel data and fixed effects

Panel data (repeated observations on the same units) introduces new moment conditions:

Fixed effects as moments:

$$\mathbb{E}[\tilde{X}_i \tilde{e}_i] = 0 \qquad \text{(within-group orthogonality)}$$

where $\tilde{X}_i = X_{it} - \bar{X}_i$ is the demeaned regressor.

Arellano-Bond (1991) GMM:

- Dynamic panels: $Y_{it} = \rho Y_{it-1} + X_{it}'\beta + \alpha_i + e_{it}$
- Lagged levels as instruments for first-differenced equation
- Growing set of moment conditions: $\mathbb{E}[\Delta e_{it} \cdot Y_{is}] = 0$ for $s \leq t - 2$
- Naturally overidentified $\Rightarrow$ GMM with efficient weighting + J-test

GMM is the workhorse estimator for dynamic panel data.

Summary

Five takeaways from Lectures 15–16:

1. **GMM is a unified estimation framework** that nests OLS, GLS, IV, and 2SLS as special cases, requiring only moment conditions $\mathbb{E}[g_i(\beta)] = 0$.
2. **Efficient GMM** uses $W = \Omega^{-1}$ to achieve the semiparametric efficiency bound. Two-step and iterated GMM achieve this in practice.
3. **The J-test** provides a natural diagnostic for overidentified models. Always report it.
4. **GMM subsumes all prior tests:** Wald, Distance, subset overidentification, and endogeneity tests are all GMM tests—valid under heteroskedasticity.
5. **Going beyond ATE:** GMM enables estimation of the MTE curve, revealing treatment effect heterogeneity that LATE conceals.